

L1 IS, System, Subsystem, IPO
Information System (IS): interrelated components that collect/input, process, store and output information to support decision making, coordination, control, analysis and visualization. System: array of components that work together to achieve goal(s). Accepts input -> processes input -> produces output. May have multiple goals. Subsystem: component of a system; can also be a system itself. Has sub-goals, transfers output to other subsystems. IPO+F: Input=data/resources, Process=transform, Output=information/product/service, Feedback=output used to control/improve future processing. Boundary: separates system from environment. Interface: point where systems/subsystems interact.

Data, Information, Knowledge, Decisions
Data: raw facts/figures (sales transactions, customer records, stock levels). Information: processed/organized data useful for a purpose (daily sales report, reorder report). Knowledge: understanding patterns/meaning from information for action (popular item/time/customer segment). Decision making: first part of problem solving; includes choice, implementation and monitoring. Structured decision: repetitive/routine; definite procedure; programmed decision. Unstructured: judgement/evaluation/insight. Semi-structured: only part has clear answer. Useful information: accurate, complete, economical, flexible, reliable, relevant, simple, timely, verifiable, accessible, secure.

L2 Business Use of IS
Digital firm: significant business relationships digitally enabled; core processes through digital networks; key assets managed digitally; time/space shifting. Six strategic objectives: 1 Operational excellence 2 New products/services/business models 3 Customer & supplier intimacy 4 Improved decision making 5 Competitive advantage 6 Survival. Operational excellence: efficiency/productivity/profitability; examples: PickMe GPS matching, Pizza Hut online order, POS billing. Business model: how company produces, delivers and sells product/service to create wealth. Customer intimacy: serving customers well -> returning customers -> revenue/profit. Supplier intimacy -> vital inputs -> lower cost. Competitive advantage: better products/services, lower price, faster response, lock-in, higher barriers. Survival: IS needed to continue business/meet industry pressure.

System Categories & Users
By application: General systems = billing, POS, HRM, payroll, inventory, invoice processing. Unique systems = hotel reservation, hospital management, student registration, patient scheduling. By number affected: Personal IS -> Workgroup IS -> Organizational IS -> Inter-organizational IS -> Global IS. By organizational level: Operational = day-to-day activities/procedures. Tactical = planning, short-term decisions. Strategic = vision, mission, long-term decisions, holistic view. By support provided: TPS -> MIS -> DSS -> SIS/ESS. Direct user: enters/uses system directly (cashier, ATM customer, rider, lecturer marking attendance). Indirect user: receives output/report through system or another person (manager, owner, HOD).

L3 IS Components
Hardware: physical devices to collect/process/store/transmit data; servers, PCs/laptops, mobiles, routers/switches, barcode scanners, printers. Software: programs/applications instruct hardware. System software: OS, utilities, antivirus, compression. Application software: word processor, spreadsheet, social media/app systems. Data: raw facts for processing; customer records, sales, employee details, inventory levels. Database: organized collection of facts/information. Data hierarchy: bit -> character -> field -> record -> file -> database. Relational model: two-dimensional tables/relations. DBMS: programs that manipulate database. People: end users, managers, system/business analysts, programmers, DBAs, IT support staff. Procedures: policies/rules/instructions; data entry, security policies, backup/recovery, approval workflows.

Data Types
Structured: predefined rows/columns, consistent, SQL-queryable, relational DB; examples customer tables, banking transactions, student registration, inventory. Semi-structured: no rigid table but tags/key-value/metadata; self-describing; easier to modify; examples JSON, XML, HTML, NoSQL document data. Unstructured: no predefined format; difficult for traditional DB; needs AI/ML/NLP/data mining; examples social media posts, images, videos, audio, voice notes, emails. Mixed/hybrid: structured tables + unstructured media/text (bank transactions + CCTV; POS tables + voice feedback).

Business Process & Process Modeling
Process: logically related tasks performed to achieve a defined outcome. Business process: interrelated activities taking input(s) and creating customer-value output. Example burger meal: activities=take order/prepare/deliver; inputs=bun, meat, labor; output=hamburger. IS support: several activities may use one IS; activity may have its own IS; activity may use several IS. Process modeling: documents how processes look, how managed/trained/compliant/improved, and how IT supports them. Importance: communication/shared understanding; end-to-end diagramming; business-IT alignment; real-time process information; better decisions; improved efficiency/effectiveness.

L4 Types of IS by Support
TPS: captures/processes/updates routine transaction data; official records; large input/output; low direct decision support; accurate/complete routine data. TPS examples: POS, ATM, airline reservation, hotel booking, payroll processing, inventory management, order processing. TPS cycle: data collection -> editing -> correction -> manipulation/calculation/sorting/summarizing -> storage/database update -> document/report production. MIS: right information to right person in right format at right time; routine/structured problems; programmed decisions; scheduled, ad-hoc, real-time reports. DSS: supports problem-specific decisions; semi/unstructured problems; complex analysis/comparisons; what-if analysis; forecasting; quantitative/model base; optimization; OLAP/pivot tables. EIS/ESS: key info for senior executives/managers from internal/external sources; overall awareness; strategic dashboards/competitor analysis.

ERP, CRM & Major Functions
Enterprise system: shares information across business functions and management levels. ERP: integrated programs managing vital operations for whole organization; shared database; consistent data across functions. CRM: manages all customer encounters (marketing, sales, service); captures every customer contact; increases retention and loyalty. Major function examples: Production=CAD, MRP, scheduling, maintenance, inventory control. Marketing/Sales=advertising, billing, POS. Finance/Accounting=AIS, finance management. HR=recruitment, talent, employee, attendance, payroll, training, grievance.

L5 Functional Area IS
Functional areas: marketing, finance, HR, operations/manufacturing, accounting, logistics, legal. They have specific responsibilities but work as an integrated system. Need for IS: operational efficiency, decision support, communication/coordination, data management, customer service, strategic planning. MKIS: supports product development, distribution, pricing, promotion effectiveness, sales forecasting. MKIS functions: understand customers, track product performance, improve ads/promotions, forecast sales. MKIS subsystems: marketing research, product development, promotion & advertising, pricing. Inputs: sales TPS, customer trends, competitor price, promotion effectiveness. Outputs: market share trends, sales forecasts. FIS: financial info for executives/stakeholders; daily decisions. Importance: cash flow management, cost tracking, resource allocation. Subsystems: profit & cost, asset management, budgeting, auditing. Outputs: statements, budget variance, investment recommendations.

HRMIS & Manufacturing MIS
HRMIS: activities related to employees/potential employees. Needs: right person-right job, reduced manual HR work, improved employee performance. HR subsystems: HR planning, recruitment, training, scheduling, payroll, outplacement. Inputs: attendance/payroll transactions, performance history, employee info, salary structure. Outputs: performance reports, training effectiveness. Manufacturing/Operations MIS: receiving raw materials, inventory tracking, scheduling, quality control. Objectives: reduce production cost/waste, improve delivery speed, maintain product quality. Subsystems: CAD, production scheduling, inventory control, JIT, quality control. Inputs: supplier info, production logs, quality test results, delivery schedules. Outputs: quality reports, stock levels. Example: stock < reorder level -> automatic supplier reorder. Decision support: Marketing=pricing/promotion/sales forecast; Finance=budgeting/investment; HR=workforce allocation; Operations=inventory & scheduling.

L6 Cloud Computing
Cloud computing: delivery of computing services (servers, storage, databases, networking, software, analytics) over internet instead of local/on-premise servers. Characteristics: on-demand access, scalable, pay-as-you-go, accessible anywhere with internet. Service models: IaaS=virtual machines/storage/networks (Amazon EC2). PaaS=platform to build/deploy apps (Google App Engine). SaaS=ready-to-use software over internet (Gmail, Zoom). Deployment models: Public cloud=shared infrastructure (AWS/Azure). Private cloud=one organization only. Hybrid cloud=public + private. Benefits: lower cost, automatic updates, backup/recovery, high availability. Risks: internet dependency, security/privacy, vendor lock-in.

AI, IoT, Mobile, Agents
AI: systems/machines performing tasks normally needing human intelligence: learning, reasoning, problem solving, language understanding, pattern recognition. AI capabilities: learning, reasoning, problem solving, perception, NLP. Benefits: automation, decision support, efficiency/productivity, personalized services. Risks: bias/fairness, privacy, job displacement, cost/data needs. IoT: network of internet-connected physical devices with sensors/software/communication to collect/share/exchange data automatically. IoT characteristics: connectivity, sensors/data collection, automation, real-time monitoring, integration. Components: devices/sensors, connectivity, data processing, user interface. IoT uses: smart homes, healthcare wearables, smart cities, manufacturing machine monitoring, agriculture smart irrigation. Risks: security/privacy, internet dependency, data management, compatibility. Mobile computing: portable access to data/apps/network while moving. Components: mobile devices, wireless communication, mobile apps. Benefits: anytime access, collaboration, remote work, location services. Risks: theft/data breach, battery, network dependency, small screen/limited power. Agent-based tech: autonomous software agents perform tasks/decisions/interactions for users/orgs. Characteristics: autonomy, reactivity, proactiveness, social ability, adaptability. Examples: chatbots, personal assistants, trading agents.

L7 IS Architecture
Technology infrastructure: hardware, software, databases, telecommunications, people, procedures configured to collect, manipulate, store and process data into information; foundation of CBIS. System architecture: conceptual model defining structure, behavior, interrelation of components; determines scalability, reliability, maintainability, security; balances performance, cost, control, complexity. Centralized processing: all processing in one location/facility; highest control; one central computer/database. Example ticket reservation central DB. Decentralized processing: remote devices at different locations; isolated; no communication between processing devices; suitable for independent units. Distributed processing: remote computers connected by network; reduces impact of one-location failure; supports availability/disaster resilience. Client-server: distributed model; server provides services/resources (DB/file/network apps); client requests services and handles UI. Two-tier: client directly connected to server/database; UI+logic often on client; simple but harder to update/scale. Three-tier/N-tier: Presentation + Application/business logic + Data; tiers updated/scaled independently; N-tier splits app tier into services. SOA: applications built from loosely coupled reusable services; service=self-contained business task (CheckCreditScore, ProcessPayment). Web services/APIs define communication contracts. Cloud-native: scalability via adding containers; resilience/self-healing; no single point of failure; scale down to optimize cost.

L8 Roles in IS Development
IS development: creating new systems or modifying existing systems; team effort among stakeholders, users, managers, IT/IS specialists; business-IT cooperation is key. Stakeholders: people who ultimately benefit from project. Users/end-users: operate/receive outputs and validate expectations. Business roles: Business Analyst=business needs, requirements, align system with goals. Systems Analyst=bridge business and IT; analyze requirements/design specifications. End users/stakeholders=requirements, feedback, validation. Technical roles: Developer/Programmer=code/functionality. DBA=design/implement/manage database. UI/UX Designer=user interface/experience. System Architect=overall structure, technologies, integration. Project roles: Project Manager=scope, time, cost, quality. Scrum Master/Agile Coach=facilitates Agile, removes obstacles. Quality/testing: QA Engineer=test plans/standards. Software Tester=functional/performance/security testing, defects. Infrastructure/support: System Administrator=servers/networks/environments. DevOps=deployment pipelines, automation, CI/CD. IT Support/Help Desk=user support/operational issues. Security/governance: Information Security Analyst=security/risk/compliance. Cybersecurity Specialist=protect threats/vulnerabilities. Emerging: Data Analyst/Scientist, AI/ML Engineer, Cloud Engineer.

End Users & SDLC Involvement
End user: person who uses/interacts with an IS or receives its output to perform work/decisions. Operational end users: perform day-to-day transactions (cashier using POS, bank staff processing transaction). Managerial end users: use reports/dashboards for decisions (branch manager, inventory manager). End-user contribution to SDLC: requirements gathering (needs/problems), analysis/design feedback (workflow/usability), testing/UAT (validate system), implementation/training (adoption), maintenance (report issues/improvements). No user involvement -> poor usability, wrong requirements, errors, resistance, low adoption. User involvement -> accurate requirements, better fit, fewer errors, acceptance.

L9 Ethics, Privacy, Digital Rights
Ethics in IS: IS influence decisions, behavior and power; failures can cause legal, reputational and social harm. Ethical concerns: Surveillance=monitoring individuals (CCTV, GPS, smart city); issue security vs personal privacy. Employee monitoring=tracking activity (email, keystrokes, productivity); issue transparency vs hidden monitoring. Algorithmic bias=bias in automated decisions; risk discrimination/unfair outcomes. Privacy: right to control personal information. Personal data: name, phone, location. Sensitive data: health records, biometric data. Regulations: GDPR (EU), CCPA/CPRA (US), PIPEDA (Canada), PDPA Sri Lanka No. 9 of 2022; PDPA inspired by GDPR. GDPR points: data minimization, integrity/confidentiality/security, accountability, access to data, right to edit inaccurate data, right to deletion, limits on automated processing, data portability. Consent: must be informed and voluntary. Good=clear opt-in. Bad=pre-ticked boxes, hidden policies. Digital rights: extension of human rights: privacy & data protection, freedom of expression, access to information/internet, freedom of assembly, security from scams/disinformation/algorithmic discrimination. Digital responsibilities: responsible use of systems; avoid misuse of personal/organizational data; privacy, data security, fair use, misinformation, digital equity/sustainability.

L10 Information Security
Information security: protection of information and elements including systems/hardware that use, store and transmit information. Information is an asset with value and must be protected. CIA triad: Confidentiality=only authorized access; Integrity=accuracy/completeness of info and processing; Availability=authorized users access info when required. Achieved by: passwords, 2FA, biometrics, encryption. I achieved by: version control, backups/recovery, checksums, digital signatures, encryption. A achieved by: offsite backup, redundancy, disaster recovery, monitoring, failover. Threat/Vulnerability/Risk: Vulnerability=weakness exploitable (weak password, shared login). Threat=anything exploiting weakness (hacker, malware, ransomware). Risk=potential impact when threat exploits vulnerability. Risk examples: Low=temporary slowdown. Medium=unauthorized staff access to test results. High=ransomware encrypting EHR/loss of critical care data. Security controls: Administrative=policies, training, access procedures. Technical=firewalls, encryption, authentication. Physical=locked server rooms, CCTV, restricted areas.

Access Control
Access control: preventing unauthorized use of resources and unauthorized manner of use; allows authorized users to access authorized resources. Authentication: verifying identity; answers Who are you? Factors: something you know (password/PIN), have (OTP phone/token/card), are (fingerprint/face/iris/voice). MFA: more than one verification factor; stronger than password alone. Authorization: verifies what actions/resources an authenticated user can access; answers What are you allowed to do? Examples: login password+OTP=authentication/MFA. Student can view marks but cannot edit=authorization. Doctor can view patient record; receptionist limited access=authorization. User responsibilities: strong passwords, do not share passwords, log out, identify phishing/suspicious links, follow policies. Organizational responsibilities: policies, training, backups, incident response, access controls, monitoring.

Scenario Reference: POS / Inventory / Reports
POS+Inventory components: Hardware=barcode scanners, computers, receipt printers, network devices. Software=POS/inventory app. Data=product codes, prices, sales, stock levels, customer purchases. Database=centralized sales/stock DB. People=cashiers, inventory managers, management, customers. Procedures=billing, scanning, reorder, backup, report generation. Direct users: cashier processing sales; inventory manager monitoring stock/reporting; customer using ATM/self-checkout/app. Indirect users: management using reports; customers receiving faster billing/receipts. Benefits: faster billing, fewer manual errors, accurate receipts, automatic stock updates, reduced shortages/over-ordering, sales trend analysis, better ordering decisions, customer satisfaction. Additional features: automatic reorder alerts, loyalty/CRM, sales analytics dashboard, supplier integration, online ordering/delivery, barcode/RFID, mobile app, backup/cloud sync. Dashboard mapping: monthly sales report -> MIS/tactical managers. 5-year profit/competitor dashboard -> EIS/ESS/strategic. Customer segment/pricing alternatives -> DSS/MKIS. Annual budget statement -> FIS. Market share/sales/customer segments -> MKIS.

Scenario Reference: Ethics / Security
MediCare/SmartWork ethics: facial recognition/biometrics -> sensitive biometric data; employee login/email/keyboard tracking -> employee monitoring; CCTV/GPS -> surveillance; AI lower ratings/rejections for group -> algorithmic bias; not informed -> consent/transparency problem. Consent violation: not informed and voluntary; unclear purpose/use/storage/access; pre-ticked/hidden policy invalid. Digital rights affected: privacy & data protection; protection from unfair automated decisions/algorithmic discrimination; security; access to information about data use. Responsible actions: inform clearly, obtain valid consent, collect only needed data, restrict access, secure biometric/health data, test/audit AI for bias, human review/appeal, explain automated decisions, update policy/training. Hospital security: shared passwords/computer left logged in=Vulnerability; ransomware/malware/hacker=Threat; records unavailable=Availability; changed/incorrect records=Integrity; data leaked/unauthorized view=Confidentiality. Controls: MFA, password policy, role-based access, auto logout/session timeout, staff training, firewall/anti-malware, encryption, backups/offsite backup, disaster recovery, monitoring, locked server rooms/CCTV.

L11 Final Exam Structure
End semester exam = 50% of module; covers LO1-LO4. Mode: computer-based using netexam.sliit.lk. Duration: 2 hours. Part A: 20 single-answer MCQs = 20 marks; 10 mixed questions (MCQ/short answer/matching/T-F/numerical/embedded cloze) = 30 marks. Part B: 2 structured real-world scenarios = 25 marks each. Module contents: IS in business organizations; components of IS; types of IS; IS in functional areas; modern IS technologies; IS architecture; roles in IS development; IS ethics, security and privacy.